This question required candidates to use the apparatus provided to describe an experiment to study how the stopping distance of a toy skier depends on its height of release. The general performance was poor. Only the more able candidates showed clearly in their answers the procedures taken, the physical quantities to be measured and the expected result based on the work-energy principle. Many candidates stated that the stopping distance would increase with the height of release, but without any graphical or mathematical elaboration. Quite a number of them did not make proper use of the long rough paper strip and the metre rule provided to perform the experiment.

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Candidates' performance was satisfactory. In (a), many candidates missed the point that if the friction F_r remained unchanged (at its maximum value) while the maximum load limit was exceeded, the braking distance d would be longer according to the equation $(1/2)mv^2 = F_r \times d$ given. In (b)(i), when explaining why it was not recommended to apply the brakes continuously, only the more able ones pointed out that the thermal energy generated would cause the temperature to reach over a few hundred degrees Celsius as mentioned in the passage. In (b)(ii), some candidates had difficulties estimating how far the vehicle would travel up the ramp by applying either the principle of energy conservation or the equations for uniformly decelerated motion. A few even did not know that the deceleration up the ramp was g sin ·.

This question tested candidates' knowledge and understanding of projectile motion in the context of a volleyball game. The overall performance was good. In (a)(i), most candidates knew that the work done on the ball became its kinetic energy. A few misinterpreted the situation and also considered the ball's gravitational potential energy in the calculation.
In (a)(ii), nearly half of the candidates computed the horizontal and vertical components of the velocity separately rather than simply using the method of conservation of mechanical energy. Weaker ones held that B was the highest point of the trajectory in (b)(i). In (c), most knew that the suggested way would result in a shorter time of flight. However, some failed to point out explicitly that this was due to a larger horizontal component of the velocity. Part (d) was well answered, although some candidates wrongly stated that a concrete floor providing larger friction was the reason.

(No Section B candidate-performance note.)

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